10.7. LOGIC PROBE

A logic probe is used to analyse and troubleshoot the logical states (0 or 1) of a digital circuit

Figure 10.27 shows a simple logic probe used to test the logical operations of basic gate circuits.



Figure 10.27 Logic probe

Red and green LEDs are used in this probe to indicate high (1) and low (0) states, respectively. Yellow LED is used to indicate narrow pulses. When the logic probe is connected to either an invalid logic level (that is a fault condition) or a tri-stated output or not connected at all, none of the LEDs will lit up. Some logic probes have a separate audible tone for each of the logical states. Logic probe can test a signal at a time. When many logic levels need to be tested at a time, a logical analyzer is used.

Uses

1. Logic probe is used to check individual gates of the TTL, DTL and CMOS ICS.
2. It is used to detect memory functions in digital equipment like computer.
3. It is also used to identify narrow pulses.

10.8. IC TESTER

Integrated circuit (shortly IC) Tester is an instrument which tests integrated chips.

IC testers are classified into three types.

1. Logic IC Tester
2. Memory IC Tester
3. Analog IC Tester

All types of ICs cannot be tested using these IC testers. Few types of ICs can be tested using these testers.

10.8.1 Testing a Logic IC Using Logic IC Tester

Figures 10.28 (a) and (b) show an AND gate and Hex INVERTER ICs being tested by an IC Tester.



(a) AND gate Using IC Tester



(b) HEX Inverter using Logic IC Tester

Figure 10.28 IC Tester

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